

BEE 208 — Question 16: Repeated DC Power Calculator

Control Statement: do...while loop | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables voltage, current, power (double) and continueChoice (char)

Step 3: Do:

Step 4: Prompt user to enter voltage and current

Step 5: Read voltage and current

Step 6: $\text{power} = \text{voltage} \times \text{current}$

Step 7: Display power in watts

Step 8: Ask user whether to perform another calculation (Y/N)

Step 9: Read continueChoice

Step 10: While continueChoice == 'Y' or 'y'

Step 11: End

2. PSEUDOCODE

```
BEGIN

DECLARE voltage, current, power AS DOUBLE

DECLARE continueChoice AS CHAR

DO

    OUTPUT 'Enter Voltage (V):'

    INPUT voltage

    OUTPUT 'Enter Current (A):'

    INPUT current

    power = voltage * current

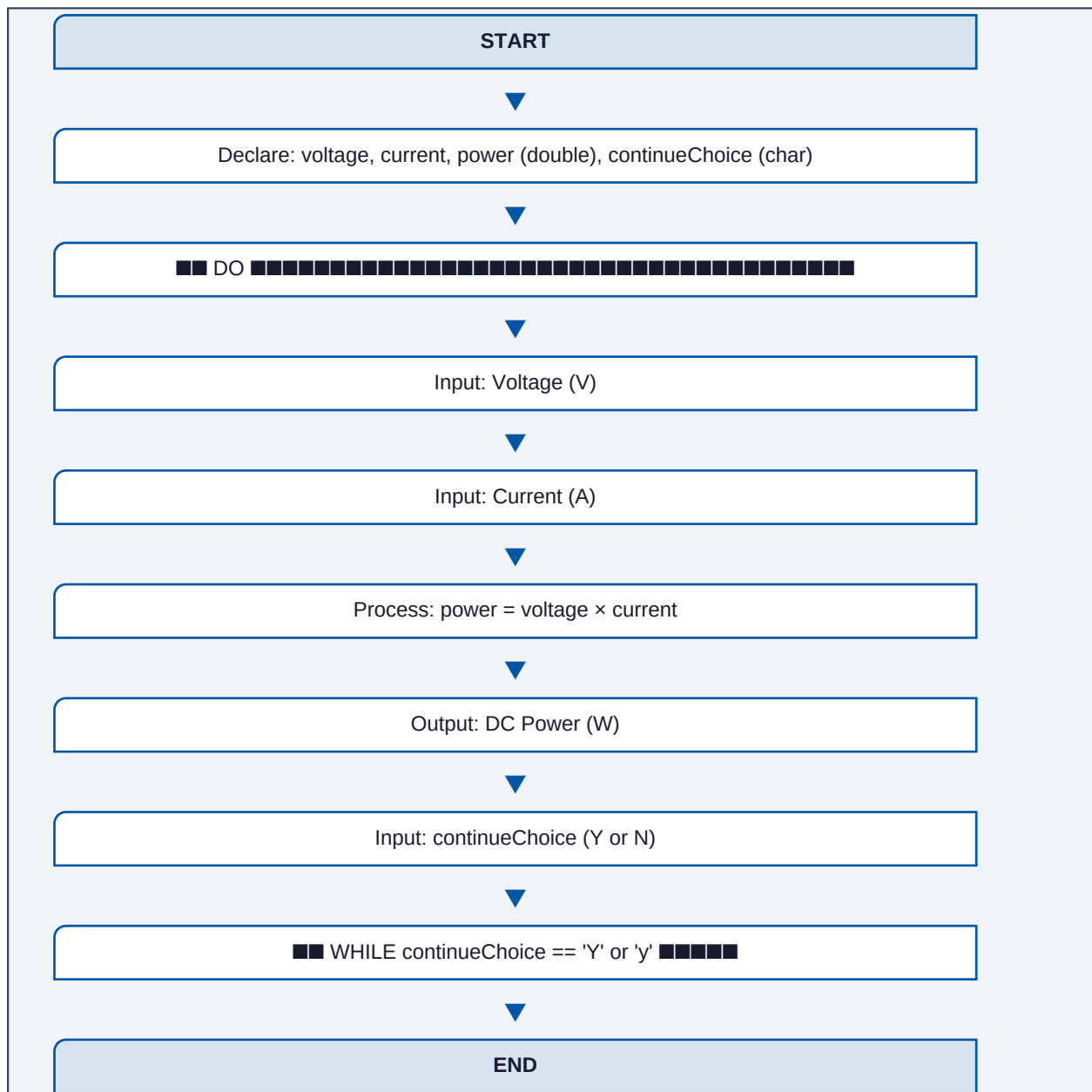
    OUTPUT 'DC Power:', power, 'W'

    OUTPUT 'Perform another calculation? (Y/N):'

    INPUT continueChoice

    WHILE continueChoice == 'Y' OR continueChoice == 'y'
```

3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	V=12V, I=5A, continue=Y; V=24V, I=3A, continue=N	Expected Output	60 W, then 72 W
Input	V=230V, I=10A, continue=N	Expected Output	2300 W